

Resonance-Avoidance Speed Control for Bridges: A Self-Optimizing Traffic–Structure Interaction Framework

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Abstract— Bridge resonance induced by moving vehicular loads often leads to excessive vibration, accelerated fatigue damage, and a reduced service life. While conventional mitigation techniques rely on costly and inflexible structural modifications like increased stiffness or damping, this research proposes a *novel Resonance-Avoidance Speed Control (RASC) framework*. By treating traffic speed as a controllable system variable, the framework creates a *self-optimizing traffic–structure interaction system* that eliminates the need for physical retrofitting.

The methodology integrates structural dynamics, moving load theory, and control logic to dynamically regulate vehicle speeds. It identifies "*resonance zones*"—specific speed intervals where the fundamental excitation frequency of moving loads coincides with the bridge's natural frequencies—and enforces speed adjustments to keep traffic within safe operating bands. Validated through numerical simulations, this system-level optimization is expected to significantly reduce dynamic amplification and decrease vibration amplitudes by 20–50%. Ultimately, this scalable and non-intrusive approach offers a low-cost alternative to traditional structural modifications, successfully improving fatigue life and introducing a new paradigm for smart, resilient infrastructure management

Keywords— Bridge Resonance, Resonance-Avoidance Speed Control (RASC), Traffic–Structure Interaction, Moving Load Theory, Dynamic Amplification Factor (DAF), Fatigue Damage Mitigation, Non-Intrusive Control

I. INTRODUCTION

The continuous movement of vehicular traffic across bridges introduces dynamic loads that can significantly impact structural integrity. When the fundamental excitation frequency of these moving loads aligns with the natural frequencies of the bridge, it induces resonance. This phenomenon leads to **excessive vibration**,

accelerated fatigue damage, and a **progressively reduced service life** for the infrastructure.

Historically, engineers have relied on conventional mitigation techniques to combat resonance, primarily through physical structural modifications such as increasing the bridge's stiffness or adding mechanical damping. While effective, these traditional retrofitting methods are **notoriously costly, inflexible, and highly intrusive**.

To address these critical limitations, this research introduces a novel **Resonance-Avoidance Speed Control (RASC) framework**, designed as a self-optimizing traffic–structure interaction system. Rather than physically altering the bridge, the fundamental philosophy of this framework is to **treat traffic speed as a controllable system variable**. Because the excitation frequency of a moving vehicle is directly tied to its speed and the bridge's span, adjusting vehicle speeds can actively prevent resonance.

The proposed methodology integrates structural dynamics, moving load theory, and advanced control logic. By evaluating the dynamic amplification factor (DAF) across various speeds, the system identifies "*resonance zones*"—specific speed intervals where the structural response exceeds safe thresholds. Once these zones are mapped, the RASC framework dynamically regulates allowable vehicle speeds, enforcing speed adjustments to ensure traffic remains within safe operating bands while avoiding restricted resonance bands.

By redefining traffic as a controllable excitation source, this research establishes a **system-level co-optimization framework that couples traffic behavior with structural health**. This scalable, non-intrusive approach is compatible with existing traffic control infrastructure and offers a low-cost alternative to traditional retrofitting. Ultimately, the RASC framework aims to **reduce vibration amplitudes by 20–50%** and significantly improve fatigue life, introducing a new paradigm in smart, resilient infrastructure design.

II. METHODOLOGY